

Two Definitions of Information in a nutshell

Shannon's definition of information links probability to information through the formula:

$$I = \log_2\left(\frac{1}{p}\right)$$

Compact codes such as the Morse code tend to associate short codes to frequent items. Ideally, based on Shannon's definition, code length L should be related to frequency F through: $L = \log_2(1/F)$.

Any code provides an upper bound of Kolmogorov complexity K . So in the absence of a better estimate, frequency provides a proxy to K :

$$K \approx \log_2(1/F)$$

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